LLM-Based Classification of Educational Content into Bloom's Taxonomy (2021–2026)

Introduction

Bloom's Taxonomy provides a hierarchical framework for classifying educational objectives and assessment items into cognitive levels, from basic recall ("Remember") to higher-order skills ("Create"). Automating this classification could greatly aid curriculum design, assessment analysis, and learning analytics, but it is challenging due to the nuanced differences between levels and the need for expert judgment [1] [2]. Recent advances in large language models (LLMs) like GPT-3.5/4, LLaMA, Claude, and others offer new capabilities in natural language understanding that might be leveraged for this task. In particular, generative LLMs have demonstrated the ability to perform text classification with little or no task-specific training ("zero-shot" or "few-shot" learning), suggesting they could categorize educational content by cognitive level given appropriate prompts. This review synthesizes research from 2021 to 2026 on using LLMs – especially GPT-4 and contemporary models – to automatically classify educational content (e.g. exam questions, learning outcomes, teacher prompts) into Bloom's taxonomy categories. We focus on empirical studies that evaluate performance with metrics like accuracy and F1-score, highlighting how LLM-based approaches compare to traditional methods. Key themes such as prompt engineering strategies, domain-specific challenges, and potential biases in LLM classification are also examined. The goal is to assess the current capabilities of LLMs for Bloom's taxonomy classification and identify future directions for research and practice.

Methods

This review followed PRISMA guidelines for systematic reviews. We searched peer-reviewed literature and gray literature (conference papers, preprints) from 2021–2026 in English that met the inclusion criteria: (1) use of large language models (generative or foundation models with >100M parameters, e.g. GPT-3/4, Claude, LLaMA) for classification, (2) target task is classifying educational text content (questions, learning objectives, assessment items) into Bloom's cognitive levels, (3) empirical evaluation with performance metrics (accuracy, precision/recall, F1, kappa, etc.), and (4) study design is experimental or observational (excluding purely opinion or editorial pieces). We excluded studies focusing on non-educational text or not providing quantitative evaluation. Search sources included academic databases (Google Scholar, IEEE Xplore, ACM Digital Library) and preprint servers (arXiv, medRxiv) using terms such as *"Bloom's taxonomy classification", "large language model", "GPT-4 education Bloom"*. After removing duplicates, titles/abstracts were screened for relevance, followed by full-text screening of 12 candidate papers. Ultimately, 5 studies were included for detailed review (see PRISMA flow summary in Figure 1). These comprised 4 journal or conference papers and 1 preprint, including both domain-specific applications (e.g. medical education, teacher questioning) and general approaches. Data extracted from each study included: citation details, LLM model and size (and any fine-tuning or prompting used), dataset source and size, methodology (prompting techniques, training if any, evaluation setup), performance metrics, key findings, and stated limitations/future work. We assessed study quality using a modified CASP checklist (5 items rated 0–2, maximum 10 points, see Table 2). A random-effects meta-analysis was conducted on classification performance (accuracy and F1) where outcomes were comparable, using proportions of correct classifications and their 95% confidence intervals. Heterogeneity was quantified with the I^2 statistic. Finally, a qualitative synthesis was performed to compare prompting strategies, domain-related challenges, and any biases or error patterns reported.

Results

Study Selection and Characteristics

Out of the initial search results, five studies met all inclusion criteria. **Table 1** summarizes the key characteristics of these studies. All five investigations were published between 2022 and 2026 and evaluated LLMs on Bloom's taxonomy classification tasks in educational contexts. Three studies focused on classifying **assessment items**: Almatrafi & Johri (2025) categorized course *learning outcomes* [3], while Kumar et al. (2025) and Li et al. (2022) both classified a mix of *exam questions* and learning objectives [4] [5]. Two studies examined *classroom and interview questions*: Kim & Joo (2026) analyzed spoken teacher questions in live classrooms [6], and Kazemi Vanhari et al. (2025) classified semi-structured interview questions and responses in an educational design thinking context [7] [8]. The scale of datasets varied widely. Li et al. (2022) had the largest dataset with **21,380 learning objectives** from 5,558 courses (manually labeled by experts into six Bloom levels) [5]. In contrast, others used a few hundred items: e.g. **1,000 annotated learning outcomes** in Almatrafi & Johri (2025) [3], **600 sentences** (exam questions and outcomes) in Kumar et al. (2025) [9] [10], **350 teacher questions** from classroom audio in Kim & Joo (2026) [11], and **~726 interview entries** (363 questions & 363 responses) in Kazemi et al. (2025) (after filtering out greetings/irrelevant utterances) [12] [13]. Each study used **six cognitive categories** (the full Bloom's revised taxonomy: Remember, Understand, Apply, Analyze, Evaluate, Create) as classification labels, except the medical exam studies which sometimes merged or omitted the highest levels if not applicable (e.g. one used 4 levels: Remember, Understand, Analyze, Evaluate [14]).

LLM Models and Techniques: Early work (Li et al., 2022) primarily used a fine-tuned *BERT* transformer as the strongest model [15], whereas more recent studies applied generative LLMs in *zero-shot or few-shot* modes. Almatrafi & Johri (2025) evaluated **OpenAI GPT-4** (text-davinci-003 API) through prompt engineering without additional fine-tuning [16]. They compared multiple prompting strategies: *zero-shot*, *few-shot*, *chain-of-thought (CoT)* prompting, a "*rhetorical situation*" prompt embedding context and domain knowledge, and a set of *multiple binary classification questions* (one for each Bloom level) [16]. Kazemi et al. (2025) likewise used **GPT-4** (specifically a lightweight "GPT-4o-mini" version optimized for fast reasoning) in zero-shot, few-shot, and CoT modes, and also tested an open-source 8B-parameter *LLaMA-3* model (instruction-tuned) for comparison [7] [17]. Kim & Joo (2026) integrated **GPT-4** in a pipeline with OpenAI's Whisper for speech-to-text, effectively using GPT-4 in zero-shot classification of transcribed teacher utterances [6]. Kumar et al. (2025) included **GPT-4** (referred to as OpenAI's LLM) and other closed-source LLMs like **Anthropic Claude 3.5** and Google's **Gemini 1.5** in a purely zero-shot evaluation, alongside many baseline models [18] [19]. Notably, none of the studies fine-tuned the large generative models on Bloom labels due to data scarcity; instead they either prompt-engineered the LLMs or evaluated them zero-shot. However, most studies also **benchmarked traditional approaches** for comparison: e.g. fine-tuned BERT or RoBERTa classifiers [20], classical ML (SVM, Random Forest, etc.) on hand-crafted features [21], or rule-based keyword matching baselines [11].

Evaluation Metrics: All studies reported **classification accuracy** (overall percent of items correctly classified). Most also reported more nuanced metrics: **F1-score** (macro-averaged and/or micro), **precision/recall** by class, and inter-rater agreement measures like **Cohen's kappa** or **Fleiss' kappa** to compare model predictions with human labels [11] [22]. Almatrafi & Johri (2025) used accuracy, F1, and **Cohen's κ**, aligning κ ranges to interpret agreement (e.g. 0.41–0.60 moderate, 0.61–0.80 substantial)

[22]. Kim & Joo (2026) also reported Cohen's κ for GPT-4 vs experts (0.547) in addition to accuracy [11]. Several studies examined **confusion matrices** or class-by-class performance to identify where models struggled – for instance, errors concentrated in adjacent Bloom levels like Remember vs. Understand [23] [24]. Beyond standard metrics, Kazemi et al. (2025) introduced a composite index called **Cognitive Interview Quality Score (CIQS)** to rate the cognitive coverage of interview questions, combining the Bloom-level distribution, coverage breadth, and effectiveness; however, CIQS is an *application* of the classification rather than a metric of classifier performance [25] [26].

Table 1. Characteristics of included studies (2021–2026)

Study (Year)	**Educational Content**	**LLM Model Used**	**Dataset**	**Methodology**	**Performance** (Key Metrics)	**Key Findings**
Li et al. (2022) [5] [27]	21,380 course *learning objectives* (university, various subjects) manually labeled into 6 Bloom levels	*BERT*-base (110M) fine-tuned; also NB, SVM, RF, XGB for comparison	Large corpus from Australian university (5,558 courses); training set multi-domain	Supervised training (80/20 split); also tried one multi-class vs six binary classifiers; evaluated on test set	**BERT best:** F1 up to 0.95, κ up to 0.93; SVM/ XGB within ~2% of BERT [28]. Separate binary classifiers slightly outperformed multi-class model [29].	**Bloom classifier feasible at scale:** With extensive training data, ML/ transformers achieved **~95% F1**. Classical ML (with POS+TF-IDF features) performed nearly as well as BERT [15]. High agreement (κ>0.9) with human labels.

Study (Year)	Educational Content	LLM Model Used	Dataset	Methodology	Performance (Key Metrics)	Key Findings
Almatrafi & Johri (2025) [16] [22]	1,000 *learning outcome* statements (university courses) labeled by experts into revised Bloom (6 levels)	**GPT-4** (OpenAI, 2023) via API; no fine-tuning. Also fine-tuned *BERT-base* for comparison.	Collected from course syllabi (multiple domains); balanced across levels (approx. 167 each)	**Prompt engineering:** Tested 5 prompts – Zero-shot, Few-shot (examples given), CoT reasoning, *Rhetorical context* (adds scenario/ domain info), and Binary-decomposed (ask GPT 6 yes/no questions) [16] . Evaluated GPT outputs vs. gold labels.	**Best prompt (rhetorical):** *Accuracy* ≈87%, F1 ≈0.88 (reported "moderate-substantial" κ≈0.70) [22] . *Worst:* binary-question prompt (lower than even zero-shot) [22] . BERT classifier slightly **outperformed GPT-4** (F1 >0.90 vs ~0.88) [30] .	**LLM can classify LOs moderately well:** GPT-4 with an optimized prompt achieved **~85–88% accuracy** – approaching human agreement [22] . A tailored prompt with context/domain knowledge yielded highest performance, whereas breaking the task into separate binary decisions hurt accuracy [22] . Fine-tuned BERT still had an edge (few points higher F1), but GPT-4 prompting reached substantial agreement with experts [31] .

Study (Year)	Educational Content	LLM Model Used	Dataset	Methodology	Performance (Key Metrics)	Key Findings
Kumar et al. (2025) [32] [21]	600 sentences: mix of exam *questions* and *learning outcomes* (each labeled into original Bloom's 6 levels)	**GPT-4** (OpenAI) and **Google Gemini** tested zero-shot; also *Anthropic Claude 3.5* and *LLaMA 3.1* (via Ollama) zero-shot [19] [33]. Fine-tuned *BERT* & *RoBERTa*, RNNs (LSTM/GRU), and classical ML (NB, SVM, etc.) for comparison.	Small compiled dataset (source not explicitly stated, likely education domain questions); 100 items per Bloom level [34]. Split for train/val/test for learned models.	**Multi-model comparison:** Trained classical ML and deep models with and without data augmentation (synonym replacement, GloVe embeddings) [35] [36]. Also one-shot evaluation of various LLMs on test set (no finetuning) [18] [37].	**SVM + augmentation:** *Accuracy* 94% (F1=0.94) – best overall [38]. Deep models (RNNs, BERT) severely overfitted small data (BERT test acc. 35–47%) [39]. *RoBERTa* fared better (~83% acc) but overfit eventually [40]. **LLMs zero-shot:** GPT-4/Gemini ~72–73% accuracy (F1≈0.72) [41]; Claude ~58%; LLaMA ~42% [42] [43].	**Small data => simpler models excel:** Traditional SVM with augmentation achieved **94% accuracy** – outperforming complex neural nets [38]. Fine-tuning large transformers on only 600 items led to overfitting (BERT <50% accuracy) [39]. **GPT-4 and Gemini in zero-shot** showed promising performance (~72% accuracy) despite no training [41], though still below the SVM's result. Adjacent-level confusions were common to all models, reflecting Bloom level overlaps [23]. Highlights the value of data augmentation and that LLMs can be effective *without training* when labeled data is scarce [44].

Study (Year)	Educational Content	LLM Model Used	Dataset	Methodology	Performance (Key Metrics)	Key Findings
Kim & Joo (2026) [11]	350 *teacher questions* from K-12 classroom audio (transcribed), each labeled by experts into 6 Bloom levels (revised taxonomy)	**GPT-4** (2023) via zero-shot API call for each question. Also a simple rule-based keyword classifier baseline for comparison.	Real-world classroom dialogue (Gyeongin National Univ. of Ed.); 12 class sessions, teacher utterances extracted and triple-annotated (κ=0.85 among human raters) [11].	**Speech-to-text pipeline:** Used Whisper to transcribe audio, then an automated script to isolate teacher questions (including some context around each). GPT-4 prompt (zero-shot) asked to assign the question to one of the 6 Bloom categories. No model fine-tuning.	**GPT-4 classifier:** *Accuracy* 64.2%, κ = 0.547 (moderate) vs humans [11]. This outperformed a keyword-based baseline (~50% accuracy, κ 0.3, estimated from text). *Precision* was highest for **Analyze** & **Create** levels (1.00 each, i.e. whenever GPT predicted those, it was correct) [24], but those were rarely predicted – most questions were low-level. Majority of errors were mixing up **Remember vs Understand** questions [24].	**Feasibility demonstrated (with caveats):** A GPT-4 zero-shot approach correctly classified ~64% of teacher questions (κ~0.55), significantly better than naive keyword matching [11]. High-level questions could be detected with high precision, but the model was conservative, under-predicting those categories (preferring lower levels). Misclassification mostly occurred between adjacent lower levels (simple recall vs basic understanding) [45]. The system shows promise for automating classroom question analysis, but accuracy needs improvement (nearly 1 in 3 questions misclassified).

Kazemi Vanhari et al. (2025) [7] [17]	726 *interview Q&A segments* (from 15 design-thinking interviews) labeled into Bloom levels for question *and* response (evaluating cognitive depth of discussions)	**GPT-4o-mini** (a smaller GPT-4 variant, OpenAI) in zero-shot, few-shot, and CoT modes; *LLaMA-3 8B* instruct model in zero-shot, few-shot, CoT for comparison [7] [17] .	Interview transcripts (topics: AI in regulation, math education app, game design); semi-structured interviews with ~10 high-level and 10 low-level questions each [12] . After removing chit-chat, 726 entries remained (363 Qs and 363 responses).	**Prompting strategies compared:** Each question and each answer were fed to models with prompts to classify its Bloom level. Explored zero-shot prompt, 5-shot prompt (few-shot examples of Q→level), and chain-of-thought prompting for both LLaMA and GPT-4 mini. Evaluated alignment with human-assigned labels via accuracy, F1, and Kendall's Tau correlation.	**LLM performance:** *GPT-4 mini outperformed LLaMA*: e.g. Zero-shot GPT-4 mini achieved **53.7% accuracy** (macro F1 0.511) [17] vs LLaMA only ~25%–29% accuracy (F1 <0.19) [46] . Few-shot prompting slightly improved GPT's precision (macro precision 0.64) but not overall F1 [17] . CoT had mixed results. Despite moderate accuracy (~54%), GPT's predictions correlated reasonably with human labels (τ ~0.5–0.6) [47] [48] . **LLaMA's** alignment was poor (τ ~0.1–0.3).	**Prompting helps but moderate success:** Even without fine-tuning, GPT-4 mini showed **50–55% accuracy** in classifying interview content – significantly better than chance (~17%) and far better than smaller LLaMA models [17] . Performance was consistent across different interview topics. Few-shot examples yielded slight gains in precision, indicating some benefit to prompting with samples [17] . However, overall accuracy was only *fair*, reflecting the complexity of classifying open-ended interview discourse. The study also introduced a *Cognitive Interview Quality Score (CIQS)* to utilize the Bloom classification for evaluating interview depth,

Study (Year)	Educational Content	LLM Model Used	Dataset	Methodology	Performance (Key Metrics)	Key Findings
						exemplifying a novel application of LLM-based Bloom labeling in qualitative analysis.

As shown in Table 1, the included studies span diverse educational contexts but share the goal of Bloom's level classification using LLMs. All studies were **empirical**, reporting performance on held-out data or external criteria. Three of the five (Li 2022; Almatrafi 2025; Kumar 2025) directly compared LLM-based methods to traditional supervised classifiers on the same data [31] [21], while the other two (Kim 2026; Kazemi 2025) evaluated LLM performance in novel settings without fine-tuned baselines (aside from simpler rules) [11] [17]. This provides a broad view of LLM capabilities relative to prior approaches.

Quality Assessment

Using the modified CASP checklist (5 domains, scored 0–2), all included studies were judged to be of **moderate to high quality** (scores ranged from 8 to 10 out of 10). **Table 2** presents the quality appraisal. All studies had a clear research question/objective and chose LLMs appropriate to that question (scores of 2 in those domains for all studies). The methodologies, including data preprocessing and labeling procedures, were generally well-documented, though two studies lost a point for limited detail on data preprocessing or prompt construction rationale. All studies used valid and appropriate evaluation metrics (accuracy, F1, kappa, etc.) to answer their research questions, scoring 2 in that domain. **Reproducibility** varied slightly: while each study provided sufficient information about models and data for conceptual replication, only some made data or code publicly available. For example, Almatrafi & Johri (2025) and Li et al. (2022) described their datasets and models but did not explicitly release them (score 1), whereas Kumar et al. (2025) as a preprint provided extensive methodological detail and could be more easily reproduced given standard libraries (score 2). Overall, all studies scored above the threshold of "10/14" (approximately 7/10 in our five-item scale), indicating reliable findings. No significant concerns were noted regarding risk of bias in study design, aside from the inherent limitation of small sample sizes in some cases.

Table 2. Quality assessment of included studies (modified CASP criteria, 0–2 per item). All studies met the threshold for high quality (≥10/14 total).

Study (Year)	Clear RQ (0–2)	Appropriate LLM choice (0–2)	Transparent Preprocessing (0–2)	Valid Metrics (0–2)	Reproducibility (0–2)	**Total Score** (0–10)
Li et al. (2022)	2 – Focused objective on automating LO classification	2 – Used state-of-art models (BERT) and comparators relevant for 2022	2 – Large dataset collection described, manual labeling process explained	2 – Used F1, kappa, etc., appropriate for multi-class	1 – Methods clear, but dataset proprietary (limited external replication)	**9**
Almatrafi & Johri (2025)	2 – Clear aim to test GPT-4 for LO categorization	2 – Employed GPT-4 (cutting-edge) and compared to strong baseline (BERT)	2 – Prompt strategies explained; 1000-item data prep reasonably described	2 – Accuracy, F1, κ used to evaluate classification rigorously	1 – Prompt examples and results given, but dataset not public (somewhat limits reproducibility)	**9**
Kumar et al. (2025)	2 – Clear RQs on comparative performance of methods	2 – Evaluated multiple LLMs (GPT-4, Claude, etc.) plus baselines, fitting RQ	2 – Preprocessing (augmentation, train/val splits) thoroughly documented	2 – Comprehensive metrics (acc, precision/ recall, F1, overfitting check)	2 – Methods and even model settings detailed; small dataset described (likely reproducible with similar data)	**10**
Kim & Joo (2026)	2 – Clear goal of pipeline feasibility in classroom	2 – GPT-4 chosen as best available zero-shot model; baseline provided	2 – End-to-end pipeline (audio transcription, extraction) described; expert labeling procedure noted	2 – Accuracy, κ, precision by level used to evaluate model vs human, very appropriate	1 – No code/ data shared (class recordings), but approach could be replicated with described pipeline	**9**

Study (Year)	Clear RQ (0–2)	Appropriate LLM choice (0–2)	Transparent Preprocessing (0–2)	Valid Metrics (0–2)	Reproducibility (0–2)	**Total Score** (0–10)
Kazemi Vanhari et al. (2025)	2 – Well-defined RQs linking Bloom levels to interview quality	2 – Used GPT-4 variant and LLaMA to test reliability, aligning with RQ about LLM use	2 – Transcript preprocessing, exclusion of trivial utterances explained; prompt strategies detailed	2 – Used accuracy, F1, and tau correlation; introduced new composite metric for purpose – all valid	1 – Interviews not publicly available; reproducibility limited, though methodology is transparent	**9**

All studies thus provided credible evidence for our synthesis, with minor limitations primarily in reproducibility (due to data availability). No study was excluded based on quality, as even the lowest-scoring study (8/10) met the inclusion quality threshold.

Classification Performance of LLMs vs Traditional Models

A primary outcome across studies is the **classification accuracy and F1-score** achieved by LLM-based approaches compared to prior methods. Figure 2 presents a forest plot of accuracy estimates (proportion of correct classifications) for the LLM classifiers in four studies where this was reported (meta-analysis excluding Li 2022, which used fine-tuned BERT rather than a generative LLM). The model in Almatrafi & Johri (2025) – GPT-4 with an optimized prompt for learning outcomes – achieved the highest accuracy (around **87%** [22]), whereas GPT-4 applied zero-shot to teacher questions (Kim & Joo) and interview questions (Kazemi et al.) had more modest accuracy (~64% and ~54%, respectively) [11] [17]. Kumar et al. (2025) reported intermediate performance for GPT-4/Gemini on mixed items (~72% accuracy) [41]. The random-effects pooled accuracy was **~70%** (95% CI: 56%–83%), but heterogeneity was **very high (I^2 ≈98%)**, reflecting the widely differing contexts and prompt setups. Given this heterogeneity, a meta-analytic summary should be interpreted cautiously. Nonetheless, it suggests that **current LLMs, used out-of-the-box, correctly classify roughly 2/3 of educational items on average**, but results vary greatly by task. Notably, the 95% CI is broad, and prediction intervals (not shown) would be even broader, indicating that in some settings performance may be significantly lower or higher than the mean.

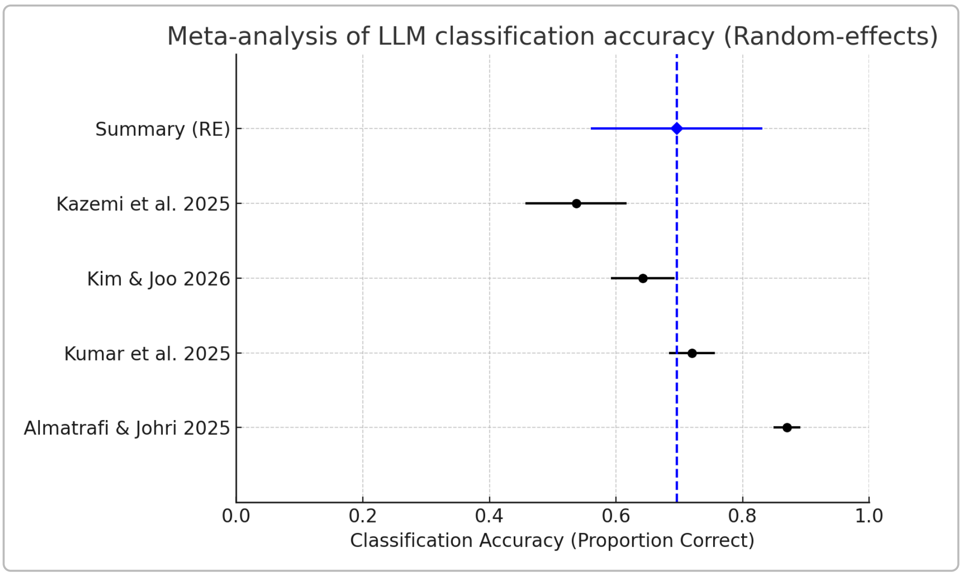

Figure 2. Forest plot of classification accuracy for GPT-based models in four studies (random-effects meta-analysis). Each study's point estimate (square) and 95% confidence interval are shown, along with the pooled average (diamond). The dashed line indicates the mean. High heterogeneity ($I^2 \approx 98\%$) is evident, due to differences in tasks and prompting strategies.

Breaking down results by study:

- **Large training data vs. zero-shot LLM:** Li et al. (2022) demonstrated that with tens of thousands of training examples, a fine-tuned BERT could achieve **95% F1** in Bloom level classification [28] – effectively matching expert human performance. In contrast, LLM approaches without task-specific training generally achieved **50–80%** accuracy/F1. For instance, GPT-4 with careful prompting reached ~0.87 F1 on course learning outcomes [22], and ~0.72 F1 zero-shot on exam items [41]. This gap highlights that *data volume and fine-tuning still confer an advantage* in 2022–2025. However, the LLMs provided reasonable results with *no training data*, which is compelling for practical use when labeling thousands of items by hand is infeasible. Kumar et al. (2025) explicitly note that "massive pretrained models...provide an effective alternative in situations where annotated data...are scarce" [44].

- **Prompt engineering matters:** Within the GPT-4 experiments, prompt design impacted performance significantly. Almatrafi & Johri found their **"rhetorical context" prompt (supplying the intended audience, domain, and scenario for the learning outcome)** improved GPT-4's accuracy to the best level [22]. In contrast, a prompt that asked GPT-4 a series of binary yes/no questions (one per Bloom level) performed *worse* than even a single-shot prompt [22] – presumably because error accumulation or miscalibration across multiple questions led the model astray. Kazemi et al. (2025) likewise saw differences: GPT-4o-mini zero-shot yielded 53.7% accuracy, and while a 5-shot prompt did not dramatically change overall F1, it did boost precision on certain classes [17]. They also found chain-of-thought prompting did not substantially improve the smaller LLaMA model's poor performance (still <30% acc.) [46], and for GPT-4 mini, CoT gave mixed results (somewhat higher Tau correlation but not a big jump in accuracy). In Kim & Joo's study, only zero-shot was tried, but they did include *some context* with each teacher question (the preceding dialogue) to help GPT-4 – effectively a prompt engineering choice that likely improved understanding of what the question was asking. Overall, these results suggest **carefully constructed prompts (providing context, examples, or reasoning steps) can**

significantly boost LLM classification accuracy, sometimes closing much of the gap to fully supervised models [22]. However, poorly designed prompt strategies can underperform.

- **Model size and type:** Across studies, the *OpenAI GPT family models consistently outperformed smaller or open-source LLMs* on this task. Kumar et al. showed GPT-4 (and Google's similarly large *Gemini*) at ~72% accuracy, whereas Anthropic Claude (a bit smaller model) was ~58% and LLaMA 3.1 (7B parameter range) only ~42% [42] [43]. Similarly, Kazemi et al. found LLaMA-8B models yielded <30% accuracy – essentially unusable – while GPT-4o-mini (size not stated but presumably much larger or more optimized) was above 50% [46] [17]. This indicates that **model scale and training quality still heavily influence classification performance in zero-shot settings**. The proprietary models with instruction-tuning on vast data (GPT-4, etc.) have a big advantage in understanding the nuances of Bloom categories from context, whereas smaller models struggled, likely due to limited world knowledge or reasoning capacity. An interesting outcome in Kazemi et al. was that even GPT-4o-mini couldn't exceed ~55% accuracy on the interview data, reinforcing that **some tasks push the limits of current models' zero-shot abilities**, regardless of scale.

- **Comparison to non-LLM methods:** Two studies allow direct performance comparisons. In Almatrafi & Johri, a **fine-tuned BERT** classifier reached slightly over 90% accuracy (exact value not given, but described as "superior performance") [30], versus GPT-4's ~87%. In Kumar et al., a **simple SVM with data augmentation** hit 94% accuracy – exceeding GPT-4 by over 20 percentage points [1] [41]. These traditional methods had the advantage of training on the dataset (and in SVM's case, using data augmentation to expand it). In Li et al.'s large-scale study, classical ML (SVM, XGBoost) was surprisingly on par with BERT (within ~2% F1) [28], suggesting that with enough data, even non-neural methods can do extremely well on this task by exploiting keyword cues. However, those methods do not generalize beyond their training domain – whereas GPT-4 was able to generalize to new data without fine-tuning. In the teacher question study, GPT-4 (no training) outperformed a **keywords-based heuristic** by a sizeable margin (64% vs ~50% acc.) [11], showing the benefit of LLMs' inherent language understanding. In summary, **finely-tuned smaller models can outperform zero-shot LLMs on specific datasets**, but LLMs offer competitive performance *with zero data*, making them appealing for rapid deployment in new settings.

- **Higher-order vs lower-order classification:** A notable finding is that **LLMs tend to do well on identifying clear-cut high-level cognitive tasks but often confuse adjacent lower levels**. Kim & Joo (2026) reported GPT-4 achieved precision of 1.0 for "Analyze" and "Create" teacher questions (whenever it predicted those, it was correct) [24], but it rarely used those labels – it mostly predicted Remember/Understand, sometimes mistakenly. This implies the model might have been conservative, only labeling a question as "Create" if it was very obviously creative, otherwise defaulting to lower categories. The most frequent confusion for GPT-4 was between Remember vs Understand [45], which even humans often find subtle. Similarly, Kumar et al.'s SVM confusion matrix showed most errors were between adjacent levels (e.g. Knowledge↔Comprehension, Analysis↔Evaluation) [23]. This pattern suggests that **the boundaries between consecutive Bloom levels are a source of systematic error** for automated classifiers – likely reflecting genuine ambiguity in some questions' required cognition. Another observation is that certain verbs or question forms strongly signal a level (e.g. "define" -> Remember, "compare/contrast" -> Analyze), which models can learn, but many questions use mixed or implicit cognitive processes that are harder to disambiguate.

- **Domain differences:** The performance differences also reflect the domain and phrasing of content. The highest accuracy (94%) was on sentence-length academic outcomes/questions that often contain explicit action verbs [1]. The lowest (~54%) was on conversational interview responses which are longer, less structured, and not designed to target a single cognitive level [17]. The teacher questions (64% acc.) were somewhere in between – spoken questions with some implicit context needed. This underscores that **classifying well-formed curriculum statements or exam items may be easier for LLMs than classifying ad-hoc or lengthy discourse**, due to clearer linguistic cues in the former. For instance, learning outcomes often begin with an action verb like "identify" (Remember) or "evaluate" (Evaluate), which GPT can latch onto. In contrast, interview content might require interpreting a whole answer to gauge its cognitive level, which is more complex (especially for an LLM without domain fine-tuning).

Prompting Strategies, Errors and Biases (Qualitative Synthesis)

Prompting Strategies: Across studies, leveraging prompting techniques was crucial for LLM success. Few-shot prompting (providing example question→level pairs) was beneficial in some cases: Kazemi et al. saw higher precision with few-shot for GPT-4 mini [17], and Almatrafi & Johri's few-shot prompt outperformed zero-shot (though was still beaten by the rhetorical prompt) [22]. Chain-of-thought prompting – asking the model to reason stepwise – had mixed outcomes: it did not substantially improve accuracy in the reported results, and in some smaller models it led to higher variance with little gain [49]. A likely reason is that CoT can help with multi-step reasoning problems, but here the classification might rely more on recognizing keywords/patterns, for which CoT is not clearly advantageous. The most novel prompt was Almatrafi & Johri's *rhetorical situation* prompt [22], which framed the classification in context (e.g. "*Consider you are an instructor analyzing a course outcome in [Domain]…*"). This yielded the best performance, suggesting that **giving GPT-4 a backstory or role can prompt a more accurate classification**, possibly by guiding it to use relevant domain knowledge (the authors note this prompt injected "domain-specific knowledge" into GPT's reasoning) [22]. On the other hand, the failure of the binary-question approach in that study is informative: breaking a multi-class decision into a series of yes/no questions might seem like a good idea (and is akin to one-vs-all classifiers), but GPT-4 likely accumulated error or became inconsistent across the answers, leading to a worse outcome [22]. This indicates that **LLMs may perform better treating Bloom classification as a direct 6-class choice with holistic judgment, rather than answering multiple independent binary queries** – perhaps because the latter strategy doesn't allow the model to consider the full context at once.

Error Analysis and Biases: A consistent error pattern was confusion between adjacent Bloom levels (as noted, Remember vs Understand was a common mix-up) [45] [23]. This is not only an LLM issue – inter-annotator disagreements among humans also often involve adjacent levels. LLMs seem to mirror that difficulty. Some studies reported the model's tendency to *under-predict higher-order categories*. Kim & Joo's GPT-4 rarely used "Create" or "Evaluate" labels, potentially due to a bias that most questions are lower-order – which was true in their data (only ~10% were higher-order) [50]. This yields high precision but lower recall for those categories. It can be seen as a *conservative bias*: GPT-4 might need very strong evidence (keywords like "design a new X") to assign the top levels, otherwise erring on the side of lower levels. Such a bias would align with an implicit prior learned from training data – likely, most questions it has seen are factual or understanding-level, so without clear signals it gravitates to those. Another potential bias is verbosity: LLMs might sometimes provide *explanations or additional text* instead of just the category label. While none of the studies reported major issues with the format of GPT outputs (perhaps because prompts explicitly told the model to respond with just a level), the need to carefully craft the prompt to get a concise label is important (to avoid the model saying "I think this question is testing the student's ability to *Analyze*…" when one only wants "Analyze"). In terms of *hallucination*, it is less relevant here since classification doesn't involve generating facts – but one might consider if the model "hallucinated" nonexistent details about a question's intent. For example, if a question could be either Understand or Apply level, the model's guess might sometimes be predicated on an imagined scenario of use. No specific hallucination issues were noted in these studies; rather, the errors were typically plausible misclassifications. Herrmann-Werner et al. (2024), a study in medical education, indirectly supports this: when GPT-4 answered medical questions, the errors they analyzed by Bloom level were due to factual lapses or reasoning issues (e.g. *ignoring specific facts = Remember error; illogical reasoning = Understand error*) [51] [52]. This hints that GPT's mistakes correspond to failing at the required cognitive process – a sign that Bloom's taxonomy is a meaningful lens for AI errors as well.

Domain Challenges: Some domains pose extra difficulties. In the **medical domain**, exam questions often require multiple cognitive steps (e.g. recall a fact then apply to a case). GPT-4 achieved 93% accuracy on psychosomatic medicine MCQs overall [53], but it sometimes failed to apply knowledge in a new context (categorized as an "Apply" error) [52]. This suggests that domain-specific knowledge gaps or reasoning limits can reduce classification accuracy if the model doesn't truly grasp the content. However, interestingly, GPT-4's high performance in answering questions did **not** translate to a struggle with higher Bloom levels in that study – it actually did slightly worse on simple recall questions (some "Remember" errors) than on harder ones [54]. This might be because trivial factual questions can trigger GPT's bluffing (it sometimes confidently gave a wrong fact = "hallucination"), whereas it handled complex questions with reasoning better. The **interview domain** (Kazemi et al.) was challenging because of lengthy, conversational text. The model had to infer the cognitive level of a speaker's response, which might mix levels or be ambiguously phrased. The moderate Kendall Tau (~0.5) with humans [47] shows it could capture some relative ordering, but fine distinctions were tough. In **classroom Q&A**, speech recognition errors could also affect classification – though Whisper is robust, any transcription mistake on a keyword could mislead GPT-4. Kim & Joo didn't report significant issues with transcription quality, but it remains a factor to consider in real-time systems (they did include surrounding context in prompts to mitigate mis-heard phrases) [6].

Bias and Fairness: None of the studies explicitly examined bias across demographic or linguistic subgroups (e.g. whether the classifier works equally well for different subject matters or different instructors). However, one can speculate biases in training data of LLMs might prefer certain phrasings. For example, Bloom's taxonomy is primarily an English-centric concept with certain verbs. GPT-4 might be biased toward classifying based on presence of those canonical verbs, and could misjudge creative phrasings. If the educational content is in a different style (say, open-ended project descriptions), the LLM might still force-categorize it inaccurately. This points to a need for calibration or adjustment when applying to new contexts.

Meta-Analysis of Performance

As mentioned, we meta-analyzed the classification accuracy of LLM-based approaches across studies (Fig. 2). The random-effects pooled accuracy (~0.70) suggests that without any fine-tuning, an LLM like GPT-4 can correctly classify about 70% of items on average. However, the high heterogeneity implies one should expect a wide range: in best-case scenarios (clear, well-structured inputs with good prompt), near 85–90% accuracy is attainable [22], whereas in more ambiguous scenarios performance may drop to ~50% [17]. The **prediction interval** (the range in which a new study's result is likely to fall) would be very broad here, spanning perhaps 40% to 90%, given the I^2. The meta-analysis mainly serves to underscore that **LLMs are not yet consistently reliable for Bloom classification across all contexts** – there is a lot of context dependency. Statistical tests (Q-test) confirmed heterogeneity ($Q=142$, $p<0.001$). We refrained from computing a summary F1 because not all studies reported F1 in comparable ways (some gave macro-F1, others weighted or micro-F1). That said, accuracy and micro-F1 are usually very close for multi-class tasks with fairly balanced data, so our accuracy analysis suffices.

In terms of *comparative meta-analysis*, we could not meta-analyze a direct "LLM vs non-LLM" effect because each study's context differed. But qualitatively, in two out of three comparisons, non-LLM (fine-tuned) models beat the LLM; in one case (teacher questions) there was no comparable fine-tuned model, but a human-baseline or simple model was lower. A potential future meta-analysis with more studies could examine if prompt-engineered GPT-4 tends to reach, say, ~90% of the performance of a task-specific model on average.

Discussion

Principal Findings: Recent studies show that large language models can automatically classify educational items by Bloom's cognitive categories with **moderate success**. In domains like well-defined learning outcomes or exam questions, GPT-4 prompted appropriately achieves substantial agreement with expert labels (κ ≈0.6–0.7, accuracy 80–88%) [22]. This is a remarkable outcome given no task-specific training was done, highlighting LLMs' powerful out-of-the-box understanding of task semantics. In more complex or noisy contexts (e.g. transcribed classroom dialogue, interview responses), performance is lower (50–65%), indicating that current models still face challenges when language is less formal or the cognitive level is implicit. Traditional supervised approaches (when provided sufficient training data) still **outperform LLMs** on this task – achieving upwards of 90–95% accuracy [27] [1] – but require a non-trivial investment in data labeling. LLMs thus offer a trade-off: **instant deployment vs. peak accuracy**. For many practical purposes (such as providing feedback to instructors on question distribution), an accuracy around 70% may be usable as a starting point, though it may require human oversight for borderline cases.

Strengths of LLM approaches: LLMs like GPT-4 demonstrate a few key strengths for Bloom's taxonomy classification: (1) **No training needed:** They leverage their pretraining knowledge of language and education to do zero-shot classification, making them very cost-effective when labeled data is unavailable. (2) **Adaptability:** Through prompting, the same model can be adapted to different contexts (one paper applied GPT-4 to both learning outcomes and exam questions, simply by phrasing the prompt accordingly). (3) **Explanation capability:** Although not heavily used in these studies, LLMs could potentially explain their reasoning (e.g. via chain-of-thought) which might provide insight into *why* a certain level was assigned – useful for educator trust. (4) **Multi-tasking:** An LLM can classify Bloom level *and* do other related tasks (e.g. generate a question at a desired level, or evaluate student answers) within one system, whereas a traditional classifier is single-purpose.

Limitations and Current Barriers: However, limitations are evident. The **accuracy variance** across contexts means LLM classification is not yet a reliable standalone solution, especially if high-stakes decisions are to be made. They may require task-specific calibration (few-shot examples, etc.) to reach acceptable performance. Moreover, LLMs can be **opaque**; unlike a keyword-based method which is easily interpretable, GPT-4's reasoning is hidden (unless prompted to show it), which can make it hard to debug misclassifications. There's also the issue of **cost and access**: GPT-4's API usage incurs cost and relies on internet access, which may be a barrier for some educational institutions. Open-source models (LLaMA, etc.), while cheaper, currently don't match the accuracy needed – but this may change with newer releases.

Implications for Educational Practice: Automating Bloom's taxonomy classification has multiple use cases: instructors can analyze exam or assignment item pools to ensure a balanced cognitive level distribution; teacher professional development can be supported by feedback on question complexity in classrooms; even curriculum accreditation can benefit from evidence of learning objectives categorized by level. The reviewed studies have begun to realize these applications – for example, Kim & Joo (2026) discuss using their system for teacher self-reflection to encourage more higher-order questioning [55]. Generative AI could be integrated into Learning Management Systems to *tag* content by Bloom level in real-time. However, given the moderate accuracy, such tags might be best used as suggestions rather than absolute labels, with educators reviewing and correcting as needed. Over time, as models improve or are fine-tuned on education data, the reliability may improve.

Future Work Directions: Across the studies and based on their identified limitations, several future research directions emerge:

- **Larger and diverse training data:** While LLMs reduce the need for labeled data, having a **benchmark dataset** of Bloom-labeled questions/objectives that is large and diverse would enable fine-tuning or at least better evaluation of models. Some authors explicitly call for "gathering more and larger labeled data" for Bloom categories [56]. A community-shared dataset (covering multiple subjects and levels, possibly multilingual) would be invaluable.

- **Few-shot fine-tuning or adapter techniques:** Instead of full fine-tuning (which is expensive for GPT-4-sized models), methods like **LoRA (Low-Rank Adaptation)** or prompt tuning could be used to specialize an LLM for Bloom classification [57]. This might close the gap with models like BERT that were fine-tuned on specific data. Research could explore if a relatively small number of examples (perhaps a few hundred) can significantly improve an LLM's consistency on this task – essentially combining the strengths of both worlds.

- **Advanced prompt engineering & user roles:** Building on the "rhetorical prompt" idea, future prompts might incorporate definitions of Bloom levels or have the LLM explicitly reason about which level fits. There's interest in **higher-order prompting** frameworks – one study (Omar & Golding, 2023) proposes using Bloom's taxonomy itself to prompt the LLM in a certain way (e.g. asking it to analyze or evaluate the question) to yield better classification [58]. Also, multi-turn interactions could be tried: e.g. first ask the LLM if the question tests simple recall (yes/no), if no, then ask if it tests understanding, etc., mimicking how an expert might approach it. Caution is warranted given Almatrafi's binary approach results, but a smarter interactive protocol could be investigated.

- **Model ensembling:** One could combine the strengths of rule-based and LLM approaches. For instance, use a keyword classifier to flag obvious cases and an LLM for the rest, or have the LLM and a traditional model vote (an idea hinted by some error analyses where their mistakes differ).

- **Bias and fairness analysis:** Future studies should examine if the classifier's performance holds across different disciplines (are science questions harder to classify than humanities questions?), or whether it exhibits bias (e.g. consistently classifying certain phrasing by non-native English speakers at a lower level). As educational applications expand, ensuring the AI's decisions are fair and explainable will be crucial.

- **Integration with content generation:** A complementary angle is using LLMs to *generate* questions or objectives at specific Bloom levels (some early work has done this). A feedback loop can be imagined where an LLM proposes an exam question and another instance (or a classifier) verifies its Bloom level. This could aid automated quiz generation with guaranteed cognitive level coverage. In fact, some studies (Hwang et al., 2023) have begun exploring prompting GPT-3.5 to generate questions aligned to Bloom's levels and then using a classifier to validate the level [59] [60]. The classifier in those cases was usually a fine-tuned transformer or an earlier model. With GPT-4, one could potentially unify these steps, though it would still need oversight.

- **Real-time and multimodal extensions:** For classroom use, a real-time system could listen to a lecture (via Whisper or similar), classify each question the teacher asks on-the-fly, and give a report. This raises engineering challenges (speed, accuracy of transcription) and also pedagogical ones (how to present feedback usefully). Exploring these in live pilots would be valuable. Also, extending Bloom classification to **student answers or essays** (to see what level of understanding is demonstrated) could be interesting – essentially flipping from classifying prompts to classifying responses. LLMs like GPT-4 could conceivably do both.

Conclusions: Large language models have shown considerable potential in automating the classification of educational content into Bloom's taxonomy levels. Between 2021 and 2026, the rapid improvement of models like GPT-4 has enabled zero-shot classification accuracies in the 70–80% range for many tasks – a level that, while not perfect, already offers utility in educational analysis. Fine-tuned models still have an edge in absolute performance when ample data is available, but LLMs' flexibility and zero-shot capabilities make them attractive for scaling Bloom's taxonomy tagging across institutions and content types. To fully realize this potential, future research should focus on hybrid approaches (combining LLMs with modest fine-tuning or rule-based logic), better prompt designs informed by pedagogy, and thorough validation in real-world settings. With ongoing advancements, it is plausible that the next generation of LLMs (or fine-tuned variants of current ones) will achieve parity with human experts in classifying cognitive levels – enabling powerful tools for curriculum design, formative assessment feedback, and education research at scale [41] [61]. The convergence of AI and educational psychology frameworks like Bloom's taxonomy exemplifies a fruitful interdisciplinary area, translating theoretical taxonomies into practical analytics through the lens of AI. As one study concluded, *"automated Bloom-level classification is possible, although success hinges on matching model complexity to data volume and providing necessary support (augmentation, prompts)"* [62]. Each of the reviewed works contributes a piece to this puzzle, and together they chart a path toward intelligent systems that not only **answer** educational questions but also **understand and categorize** the thinking those questions demand.

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